

The Four-Question Checklist

Before you call it fixed — StepChangeAnalysis.com

A fishbone diagram or 5 Whys chain gives you a **candidate** cause. Most root cause analysis stops there — diagram drawn, causes listed, action plan signed off. In practice the chain breaks after the first link far more often than it should. Four separate checks need to pass before an intervention is genuinely done, not just completed. Work through them in order — each one assumes the previous check has passed.

1

Right cause?

Is this actually the root cause — or a comfortable stopping point?

- The cause you found is a system feature, not a person — it would produce the same outcome with someone else in that situation.
- You've gone past Structure to Mindset — the belief or assumption that keeps the structure in place.
- You checked for psychological safety before trusting the answer — fear produces a sanitised cause, not the real one.

Go deeper: stepchangeanalysis.com/making-the-invisible-visible.html (Iceberg & Mindsets) · stepchangeanalysis.com/root-cause-analysis.html (fishbone, 5 Whys, fault tree)

2

Right intervention?

Is the fix pitched at the level the cause actually sits at?

- You've named the level: Level 1 (fix the output), Level 2 (fix the process), or Level 3 (fix the system).
- If the cause is Level 3, the fix is Level 3 — not a Level 1 patch dressed up as a solution.
- You've checked the fix is necessary *and* sufficient — finding the cause doesn't guarantee the fix will hold.

Go deeper: stepchangeanalysis.com/joiner-levels-of-fix.html · stepchangeanalysis.com/necessary-but-not-sufficient.html

3

Right measure?

Are you tracking the measure that will actually tell you?

- You're measuring the outcome, not just whether the action was completed.
- If the outcome is years away, you have a leading indicator that moves before the outcome does — not a vanity metric.
- The measure is paired with a balancing measure, so it can't be gamed by optimising the number instead of the purpose.
- You've checked whether a genuine positive outlier (a Bright Spot) already exists somewhere in the system.

Go deeper: stepchangeanalysis.com/types-of-measures.html · stepchangeanalysis.com/leading-indicators.html · stepchangeanalysis.com/measurement-gaming.html · stepchangeanalysis.com/bright-spots.html

4

Right analysis — did it work?

Did the outcome measure actually change, or are you assuming it did?

- You pre-specified the prediction before implementing the fix — expected change, by when, at what confidence.
- You ran the outcome series through Bootstrap CUSUM rather than eyeballing a before/after chart.
- You checked the result isn't normal noise being mistaken for a signal, or a signal being dismissed as noise.
- If the result is flat, you've ruled out tampering — reacting to common cause variation as if it were special cause.

Go deeper: stepchangeanalysis.com/interpreting-stepchange-analyzer-output.html ·

stepchangeanalysis.com/tampering-and-impatience.html · stepchangeanalysis.com/false-alarms-in-performance-charts.html

Method: Bootstrap CUSUM step-change analysis. Taylor (2000), building on Page (1954), Hinkley (1971), and Efron & Tibshirani (1993).

Frameworks: Joiner, *Fourth Generation Management* (1994) · Goldratt & Cox, *The Goal* (1984) · Senge, *The Fifth Discipline* (1990) · Deming, *Out of the Crisis* (1982).

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